

Tyler Rafferty

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EDUCATION

University of Massachusetts Amherst

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 3.83 / 4.0

Honors & Activities: MassAI Club, HackUMass 2024, Dean's List (Fall 2024 & Spring 2025 & Fall 2025)

Relevant Coursework: Algorithms, Database Management, Computational Intelligence, Linear Algebra, Programming Methodology

Amherst, MA

Expected Dec 2026

PROFESSIONAL EXPERIENCE

Fidelity Investments

Software Engineering Intern

Merrimack, NH

June 2026 – August 2026

- Built an automated API compliance monitoring system in **Python/Django**, leveraging **GitHub Copilot** to accelerate development, that issued requests with invalid authentication tokens and flagged non-compliant responses, helping ensure the security of **1,600+ APIs** across Fidelity
- Designed and developed an **Angular/TypeScript** dashboard displaying endpoint ownership, test results, status codes, and error reporting
- Developed backend scanning and data persistence services using **Django** and **PostgreSQL**, used **Snowflake** to retrieve information, deployed the application internally, and managed source code with **Git**
- Collaborated in an **Agile** environment with daily standups and two-week sprints, tracked work in **Jira** and **Confluence**, and presented the project and results to the **SVP, Head of Technology**

Federal Reserve Bank of Boston

Software Engineering Intern

Boston, MA

June 2025 – August 2025

- Expanded partner testing capabilities by developing a full-stack feature (React + Spring Boot) that introduced randomized input fields, improving testing depth and reliability for **100+** financial institutions using the platform
- Strengthened production confidence by engineering unit and integration test suites (Jest, JUnit) that increased code coverage across the new feature, catching edge cases before deployment
- Accelerated feature delivery by collaborating with product owners and senior engineers in a distributed agile environment, refining requirements to align functionality with FedNow's large-scale testing strategy

PROJECTS

CustomGuess | Next.js, Golang, PostgreSQL, WebSocket, AWS, Docker

CustomGuess GitHub

- Built a real-time multiplayer deduction game using Next.js 15, Go, and PostgreSQL, supporting live gameplay between concurrent players through a central WebSocket pub-sub hub, with **2,000+ registered users** and **30,000+ games played**
- Sustained **1,000+ concurrent WebSocket connections** with p95 message delivery latency of **2ms** on a single t3.micro EC2 instance, validated under load testing with k6
- Deployed a containerized Go backend to AWS EC2 (Terraform-provisioned) and Next.js frontend to Vercel, with a GitHub Actions CI/CD pipeline running build/test gates on every PR and auto-deploying the backend to ECR/EC2 on merge

FinanceApp | TypeScript, AWS, Terraform

FinanceApp GitHub

- Architecting a serverless, event-driven personal finance tracker on **AWS** (API Gateway, Lambda, RDS Postgres, Cognito, SQS, EventBridge), using Claude to iterate on infrastructure and schema design decisions
- Modeled a multi-user relational schema (users, accounts, categories, transactions) in **TypeScript** with Drizzle ORM, enforcing per-user data isolation via foreign-key scoping instead of a shared-tenant model
- Authored **Terraform** IaC provisioning the VPC and RDS Postgres instance, structuring modules by concern (network, database, auth, api, async) for incremental, independently testable delivery
- Designed an event-driven pipeline (SQS + EventBridge + Lambda consumers) for async weekly digest emails and Plaid Sandbox transaction sync

CampusBikes | Next.js, React, Spring Boot, PostgreSQL, Docker

CampusBikes GitHub

- Designed and built a full-stack bike and scooter rental platform with a small team, enabling UMass students to post, browse, and rent rides through a streamlined listing and reservation flow
- Implemented JWT-based authentication and role-protected API endpoints in Spring Boot, ensuring only verified UMass community members could access the platform
- Containerized the three-service architecture (React frontend, Spring Boot API, PostgreSQL) with Docker Compose, reducing environment setup from manual multi-step configuration to a single command

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Golang, Python, Java, HTML, CSS, SQL, NoSQL

Frameworks & Libraries: React, Angular, Node.js, Express, Django, Spring Boot

Cloud & Tools: AWS (EC2, Lambda, API Gateway, RDS), Terraform, Docker, PostgreSQL, MongoDB Atlas, CI/CD, Git, GitHub, LaTeX, Nginx, Figma, Linux

Concepts & Skills: RESTful API development, Microservices, Event-Driven Architecture, Agile Development

Certificates: Docker Foundations Professional Certificate